Retail Sales Analysis & Prediction

Project Report | Data Analysis, Visualization & Machine Learning

1. Project Overview

This project analyzes retail sales data to explore sales and profit patterns across regions, product categories, time periods, and shipping performance. It also develops machine-learning models to predict sales and compares an original Random Forest model with an improved version.

2. Main Analysis & Visualizations

- Regional sales distribution and regional profit performance.
- Sales and profit comparisons across product categories.
- Monthly sales performance and a monthly sales heatmap.
- Shipping performance analysis.
- Actual versus predicted sales comparison.
- Top feature importances and original-versus-improved model performance.
- Three-dimensional visualizations of sales by region and category, and profit by category.

3. Machine-Learning Model Results

Metric	Original Random Forest	Improved Random Forest
MAE	215.75	203.35
RMSE	640.24	627.11
R^2 Score	0.2148	0.2467

The improved model performs slightly better: MAE and RMSE decreased, while the R^2 score increased. However, the R^2 score is still relatively modest, so the model's predictions should be treated as estimates rather than highly accurate forecasts.

4. Important Features

The leading features in the model's feature-importance results include Quantity, Order_Month, Order_Year, Sub-Category_Machines, Order_DayOfWeek, and Discount. These features contributed more to the model's predictions than the other features in the reported ranking. Feature importance indicates model reliance, not causation.

5. Conclusion

The project combines exploratory data analysis, visual storytelling, and predictive modeling in one retail-sales workflow. The improved Random Forest model shows a small measurable gain over the original model. Further work could test additional algorithms, improve feature engineering, and validate performance on more data.

6. Project Files

The Jupyter Notebook contains the analysis and model workflow. Visualization images belong in the Images folder, and this PDF belongs in the Reports folder. Add the final GitHub repository link here when the repository

documentation is complete.